

SHYAM SUNDAR SINGH

AI / ML Engineer | Machine Learning | Generative AI | Computer Vision

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SUMMARY

Data Science / ML Engineer with 2+ years of production experience designing, training, and deploying machine learning models, including applied work in deep learning, computer vision, and Generative AI (CLIP, HuggingFace Transformers, meta-learning, few-shot and zero-shot classification). Skilled across the full ML lifecycle — data preprocessing and feature engineering, model development in TensorFlow and PyTorch, and deployment of production-ready Flask REST APIs on AWS. Comfortable working with cross-functional teams to translate ML models into measurable business outcomes and looking to bring this experience to an AI/ML Engineer role building GenAI-enabled, cloud-deployed applications.

CORE SKILLS (ALIGNED TO ROLE REQUIREMENTS)

- Machine Learning & Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, neural network design and tuning
- Generative AI / NLP / LLMs: HuggingFace Transformers, CLIP (vision-language model), prompt engineering, meta-learning (Prototypical Networks, MAML), zero-shot and few-shot inference
- Computer Vision: OpenCV, YOLOv4, object detection, image processing, real-time video pipelines
- MLOps / Production Deployment: Flask REST APIs in production, AWS, Git/GitHub, Postman, data pipeline standardization
- Data Science & Analytics: Predictive modeling, feature engineering, Pandas, NumPy, SQL, MySQL, MongoDB, Tableau, Matplotlib, Seaborn, Plotly
- Languages: Python, SQL, Java, JavaScript

WORK EXPERIENCE

Data Science Engineer

Feb 2024 – Present

VIVAX.IN

- Designed and deployed classification and regression models for customer behavior analysis, increasing prediction accuracy by 18%.
- Deployed predictive ML models via production Flask REST APIs serving 500+ daily requests, handling the full pipeline from model training to live inference.
- Built a customer churn prediction model (Random Forest) achieving 82% accuracy, contributing to a 12% improvement in customer retention.
- Standardized data cleaning and transformation pipelines across recurring projects, cutting feature noise by 35% and improving model reliability.
- Automated daily reporting workflows with Python scripts, reducing manual processing time by 70% (~10 hours/week), demonstrating ability to operationalize ML/AI systems for business efficiency.

Software Development Intern

Sep 2022 – Nov 2022

NBLIK

- Redesigned an analytics dashboard UI, increasing average user session duration by 40%.
- Built an equipment monitoring module for production processes, reducing unplanned downtime by 20% and improving customer satisfaction by 15%.
- Refactored dashboard modules to align with brand guidelines, improving brand consistency scores by 25% in user surveys.

RELEVANT PROJECTS

AI Trust Level Prediction System

Apr 2026

Python, Scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy

- Designed a predictive analytics pipeline to estimate AI Trust Level (%) from AI perception datasets.
- Conducted data preprocessing, feature engineering, and missing value treatment to improve model performance.
- Evaluated multiple regression algorithms including Linear Models, Random Forest, Gradient Boosting, and Boosted Tree ensembles.
- Performed model validation using K-Fold Cross Validation and optimized hyperparameters through RandomizedSearchCV.
- Built an ensemble learning solution combining CatBoost, XGBoost, and LightGBM to improve predictive performance.
- Automated prediction generation and submission file creation for competition-ready deployment.

Low-Resource Image and Text Classification using Meta-Learning

Mar 2026

Python, PyTorch, HuggingFace Transformers, CLIP, Scikit-learn

- Designed a meta-learning pipeline (Prototypical Networks, MAML) reaching 80%+ 5-way accuracy with only 5 labeled samples per class.
- Applied CLIP for zero-shot image recognition, classifying unseen categories from text descriptions alone — direct experience with vision-language GenAI models.
- Ran prompt-based inference using HuggingFace Transformers for low-resource text classification, cutting annotation requirements by 90% vs. fully supervised training.
- Benchmarked zero-shot vs. few-shot vs. supervised approaches across datasets with fewer than 50 samples, documenting accuracy/generalization trade-offs.

Real-Time Face Detection and Counting System

May 2024

Python, YOLOv4, OpenCV

- Fine-tuned a YOLOv4 model on a custom dataset of 5,000+ annotated images, achieving 92% mAP with 30+ FPS for real-time face detection.
- Developed a real-time face detection and counting system using Python and OpenCV, capable of processing live webcam and video streams with high accuracy.
- Implemented image preprocessing, data augmentation, and annotation techniques to improve model robustness under varying lighting conditions, camera angles, and occlusions.
- Optimized the inference pipeline to reduce latency and improve detection speed, enabling smooth deployment on real-time video feeds.
- Evaluated model performance using mAP, Precision, Recall, and F1-Score, ensuring reliable and accurate face detection across diverse test scenarios.

Personalized Loan Demand Prediction — MachineHack Hackathon

July 2025

Python, XGBoost, LightGBM, CatBoost, Scikit-learn

- Cleaned and transformed 10,000+ customer records, handling 95% missing values and cutting feature noise by 30%.
- Tuned 4 ensemble models via GridSearchCV/RandomizedSearchCV across 100+ parameter combinations; ranked top 15% among 500+ participants.

EDUCATION

B.Tech, Computer Science and Engineering

May 2024

Amity University, Lucknow Campus | Lucknow, India

CERTIFICATIONS & ACHIEVEMENTS

- Git and GitHub — Udemy (Certificate of Completion)
- MachineHack Hackathon — Ranked top 15% among 500+ participants, Jan 2025
- Internship Certificate — NBLIK, via Internshala, Nov 2022
- LeetCode — 170+ problems solved across Data Structures and Algorithms (Username: shyam0056)